The Thin Lens Equation

> **SYNOPSIS**
>
> One equation locates the image, a second says how big it is and which way up, and a short table of signs makes both of them work for every lens in the module — converging or diverging, image real or virtual. The equations are two lines. The signs are the note.

BEFORE YOU START — PREDICT

The same 3 cm object is put in front of the same 10 cm converging lens three times: at **50 cm**, at **5 cm**, and at **20 cm**. One of those three produces an image that is *upright*. Which one, and why that one?

PERFORMANCE EXPECTATIONS

1. **Apply** $1/d_o + 1/d_i = 1/f$ to locate an image.
2. **Apply** $m = -d_i/d_o$ to get its size and orientation.
3. **Read** the sign of each quantity correctly, and say what each sign means physically.
4. **Recognise** from a negative d_i that the image is virtual, and say which side of the lens it is on.

2. Thin lens equation

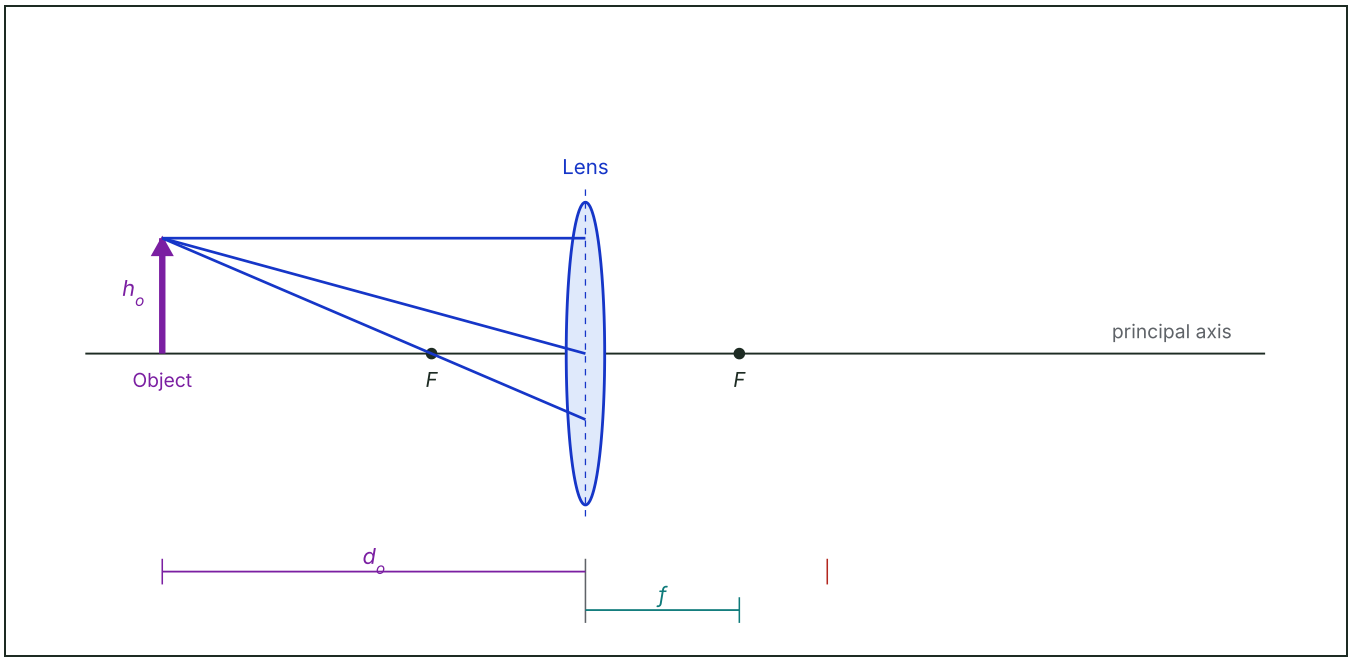


Figure 1. Converging lens, object outside the focal point. The three rays are not placed by eye: the image sits where $1/d_o + 1/d_i = 1/f$ puts it, and all three cross there because the arithmetic forces them to.

Your job on Figure 1. The object, the lens, both focal points and the three incident rays are drawn for you. Carry each ray through the lens by its own rule, mark where they cross, and draw the image there. Then check the position against $1/d_o + 1/d_i = 1/f$ — and on Figure 2, trace the two refracted rays *backwards* to find the virtual image.

$$1/d_o + 1/d_i = 1/f$$

and the magnification — image height over object height — is

$$m = -d_i / d_o$$

The signs

Quantity	Positive means	Negative means
d_o — object distance	real object	virtual object
d_i — image distance	real image	virtual image
f — focal length	converging lens	diverging lens
m — magnification	upright	inverted

Four rows, and every question in this note is answered by reading them in the right order: signs in, arithmetic, signs out.

Worked example

Find the location, orientation and magnification of the image of a **3 cm high object** placed at **(a) 50 cm**, **(b) 5 cm** and **(c) 20 cm** in front of a convex lens of **10 cm focal length**.

1. Work all three before you open the solution. The arithmetic is four keystrokes each; the sign of the answer is the part worth checking.

What about a concave lens?

Nothing new. The same two equations, with f entered as a negative number.

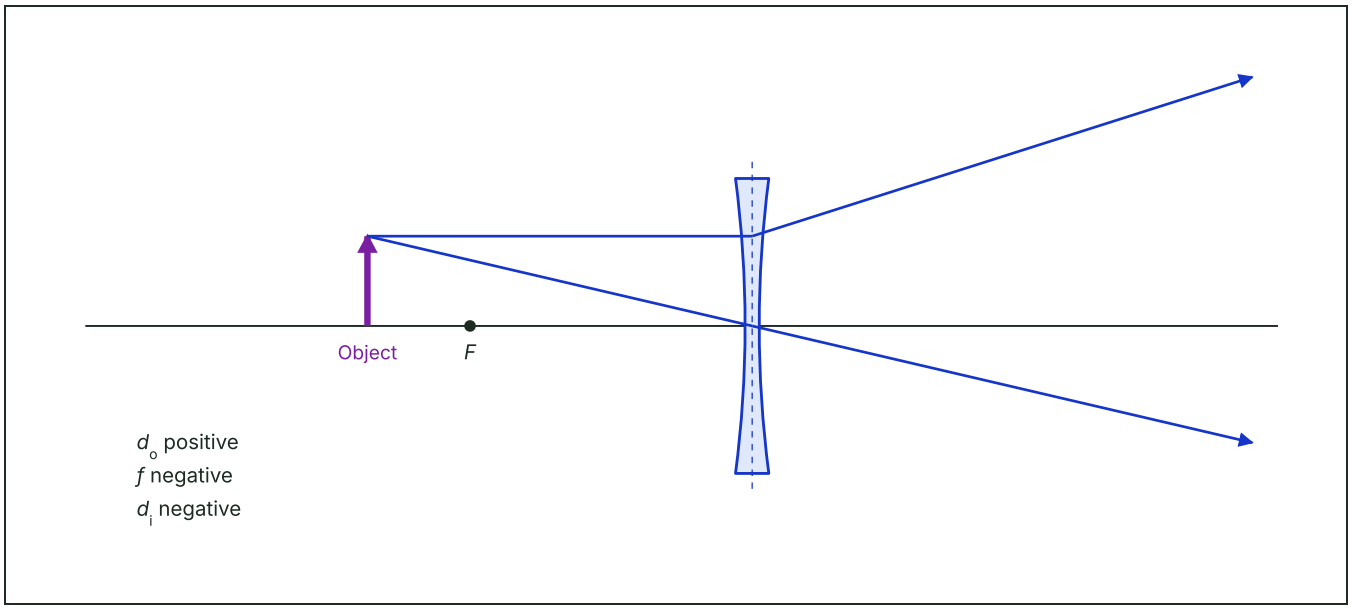


Figure 2. A concave lens, worked the same way. Nothing about the equation changes — only the signs you feed it. $f < 0$ comes out as $d_i < 0$, and a negative image distance means the image is on the **same side as the object**: virtual, upright, and smaller.

THE PUNCHLINE

$$1/d_o + 1/d_i = 1/f \quad m = -d_i/d_o$$

Back to the prediction: the upright image is **(b)**, at 5 cm — the one case where the object sits *inside* the focal length. That is not a special equation; it is the same equation returning a negative d_i.

Looking ahead

So far the image has been found by arithmetic. **Note 7** finds it with a ruler instead: three rays whose behaviour at a lens you can state in one line each, and which cross exactly where the equation says they will.

2. For that same object and lens, what is the magnification m?

3. A calculation returns $d_i = -8$ cm. What does the minus sign tell you?

○ The image is inverted. ○ The image is virtual, on the same side of the lens as the object. ○ The lens must be diverging. ○ The object was placed at a negative distance.

4. An object is 15 cm in front of a **diverging** lens of focal length −30 cm. Find d_i, in cm.

cm

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.